

The noun list, two years on

Where the SEN dataset stopped, nine things wrong with it, the sources that appeared while it sat, and the one analysis that turns a word list into a game.

Reviewed 27 Aug 2026 Source: ListOfNouns.ipynb · Word Chain.docx · nouns240123.csv

All figures recomputed from your files

WHERE YOU STOPPED

A complete extraction, an unstarted classification

The pipeline ran end to end and the output is intact. What never started is the part that would make the dataset useful: sorting nouns by who can be expected to know them.

41,466 NOUNS SHIPPED	14,884 NAMES SHIPPED	37.7% NOUNS WITH A WIKIPEDIA GLOSS	0 NOUNS CLASSIFIED BY TIER
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STAGE	NOUNS	NAMES
WordNet 3.0 noun lemmas	146,347	—
Split on capital letter	103,360	42,987
Drop 1-character entries	103,313	—
Deduplicate lemmas	78,565	40,433
Drop digits, periods, slashes, apostrophes	77,673	—
Drop underscores and hyphens	41,466	14,884

Two TODO cells sit unexecuted at the bottom of the notebook — *check if 11K and 6K8 are included* and *classify by specialty and age or education* — followed by an abandoned spaCy and scispaCy exploration that returned zero medical entities. In `Word Chain.docx` you drafted a ten-class LLM prompt (ABR / NAM / GEO / GNO / 05Y / 10Y / 50Y / BIO / MED / TEC) and never ran it. That prompt is still the right destination. It is not the right method — see finding 7.

HANDLE FIRST

Word Chain.docx contains a live Wikimedia API client secret and a non-expiring access token in plain text, along with a server IP and username. The document is stored in a cloud project. Revoke and reissue the credentials at api.wikimedia.org, then delete that block from the file. The token's payload decodes to an expiry in the year 2075, so it is effectively permanent until revoked.

Nine findings, recomputed from your own files

Findings are tagged by kind: **DATA** a defect in the shipped list, **METHOD** a technique that cannot produce what it was asked for, **SCOPE** a design decision worth revisiting rather than an error.

DATA

1 · The source dictionary is frozen at 2006

NLTK ships Princeton WordNet 3.0, released in 2006 and never updated. Your notebook's vintage was already 18 years at extraction; it is 20 now. I ran a 68-word spot check of ordinary post-2006 vocabulary against your list.

28 of 68 missing – smartphone, selfie, podcast, emoji, hashtag, chatbot, malware, bitcoin, blockchain, influencer, wifi, app, vape, quinoa, pilates, hoodie, earbud, heatwave, microplastic, telehealth ...

METHOD

2 · "Contains a capital letter" is the wrong proper-noun test

It is wrong in both directions. It files acronyms that are not names at all (pH , mRNA , AARP) into names.csv, and it leaves genuine proper nouns in the noun list wherever WordNet lemmatised them lowercase. WordNet already marks proper nouns structurally, via `instance_hypernyms()` .

693 entries in nouns240123.csv are WordNet instances – amazon, andromeda, atlas, babel, bacon, badlands, balboa, baedeker, anjou, astrakhan ...

DATA

3 · Plural contamination is systematic, not anecdotal

The notebook hand-excludes exactly two words, shorts and pants , and lists six more as "typical mistakes." The actual scale is two orders of magnitude larger. Counting only entries ending in -s whose own singular also appears in the list:

452 entries – acoustics, aesthetics, affairs, arms, arts, assets, blues, bowels, bleachers, belongings, braces, auspices, antipodes ...

DATA

4 · British and American spellings are counted as separate nouns

For chain play these are one word with two skins; for any future frequency ranking they split a single word's count in half. Neither the dataset nor the notebook records which is which.

215 -isation/-ization pairs · 43 -our/-or pairs – amortisation, authorisation, atomisation, arbour, ardour, armour, behaviour, candour, colour ...

DATA

5 · The CSV silently loses two words on every read

Pandas parses the strings `nan` and `null` as missing values. Both are real English nouns and both are in your list. Any downstream `read_csv` without `keep_default_na=False` drops them and may crash on a string operation, which is what you would have hit on resuming.

```
pd.read_csv('nouns240123.csv', keep_default_na=False, na_values=[])
```

METHOD

6 · The spaCy NER experiment could not have worked

You joined 41,466 context-free nouns into one string and asked a named-entity recogniser to label them. NER models infer entity type from surrounding sentence structure – capitalisation, determiners, verbs, position. A comma-delimited word dump supplies none of that, so the model is guessing from orthography alone. You spotted the symptom in the docx ("*plenty of medical and biological terms as geographical*") and correctly doubted the result; the diagnosis is that the input was invalid, not the model. The same applies to the scispaCy run that returned zero entities.

METHOD

7 · The classification you postponed was already inside WordNet

Every WordNet noun synset carries a *lexicographer tile* — a 26-way supersense assigned by hand by the Princeton lexicographers. It is free, deterministic, and needs no model. It maps almost directly onto your BIO / MED / TEC / GEO buckets, and it separates the specialist vocabulary that inflates your list.

person 6,130 · artifact 5,998 · act 3,665 · state 2,938 · attribute 2,804 · communication 2,765 · animal 2,424 · plant 2,349 · substance 2,227 · cognition 1,777 · food 1,075 · body 1,045 ...

METHOD

8 · Wikipedia measures notability, not wordhood

The enrichment pass cost thousands of API calls and reached 37.7% coverage. That number is not a validity signal. *Aardvark* has an article because aardvarks are things; *rarity*, *clemency* and *jauntiness* do not, and they are better word-chain words than *aardvark*. Used as a filter it would delete precisely the common abstract nouns you want and keep the taxonomic Latin you don't. Keep the summaries as flavour text for a game UI; do not let them near the inclusion decision.

SCOPE

9 · Discarding compounds threw away 46% of the corpus undocumented

35,247 underscore entries and 3,189 hyphenated entries were dropped for clarity. That is a rules choice — many word-chain traditions accept *ice cream* → *milk* — and it is invisible to anyone who receives only the CSV. It also swept out single-token-in-speech compounds that WordNet happens to store with a separator, such as `sou1_mate`, which the notebook itself flags. Keep the discards as a labelled column rather than deleting them.

NEW ANALYSIS

The letter economy

This is the analysis the project never ran, and it is the one that turns a word list into a game design. In Word Chain every noun is an *edge*: it consumes its first letter and produces its

last. So each letter has a supply — words that start with it — and a demand — words that end with it. Where supply is far below demand, the letter is a trap.

Ranked by supply ÷ demand across your 41,466 nouns:

	WORDS ENDING IN	WORDS STARTING WITH	RATIO	SUPPLY AGAINST DEMAND
Y	3,619	119	0.03	
N	4,742	937	0.20	
X	197	40	0.20	
E	6,775	1,546	0.23	
R	4,561	1,781	0.39	
K	870	484	0.56	
T	3,619	2,347	0.65	
G	2,073	1,452	0.70	
L	1,537	1,449	0.94	
S	4,403	4,794	1.09	
A	2,116	2,814	1.33	
D	1,611	2,201	1.37	
M	1,742	2,473	1.42	

O	582	937	1.61	
H	989	1,829	1.85	
Z	52	112	2.15	
U	123	532	4.33	
W	215	946	4.40	
I	262	1,336	5.10	
P	635	3,719	5.86	
F	186	1,655	8.90	
C	359	4,239	11.81	
B	179	2,475	13.83	
J	6	343	57.2	
V	10	685	68.5	
Q	3	221	73.7	

Y is the kill move. 3,619 nouns end in y and 119 begin with it — and only 46 of those 119 are common enough to have a Wikipedia article. In practice the entire counter-supply a human player can reach is about forty words: yam, yak, yard, yarn, year, yeast, yellow, yen, yew, yoga, yogurt, yolk, youth, yo-yo... A player who knows that list wins; a player who doesn't loses to -y every time.

The mirror image is that *j*, *v* and *q* are the safest possible **endings** — almost nothing ends there, so those letters never come back at you. Conversely they are terrible things to be handed, because so few words end in them that the chain barely branches.

None of this is in the dataset. Three derived columns — `start_supply`, `end_demand`, `trap_score` — would make it the first noun list that knows anything about the game it was built for.

WHAT APPEARED WHILE THE PROJECT SLEPT

Sources that did not exist, or did not matter, in January 2024

Open English WordNet 2025 edition · released 31 Dec 2024 · annual

The living successor to Princeton WordNet 3.0, maintained by the Global WordNet Association. Ships as GWA XML, RDF, JSON and legacy WNDB. Crucially, the 2025 edition **moved proper nouns out into a separate resource, Open English Namenet**, and offers a "2025+" build that keeps curated proper nouns in — which is exactly the nouns/names split you hand-rolled with a capital-letter regex.

USE FOR replacing the base extraction and finding 2 outright.

The wn Python library v1.0.0 · Feb 2026

A modern replacement for `nltk.corpus.wordnet`. It downloads and queries any WordNet edition — including OEWN 2025 — from a local SQLite index, instead of pinning you to whatever NLTK vendored a decade ago.

USE FOR the extraction layer of the rebuilt notebook.

▼ [WordNet](#) / [1.0.0](#) · [releases](#) · [changelog](#) · [license](#) · [about](#)

Machine-readable English Wiktionary as JSONL: **964,187 noun senses** and 274,572 proper-name senses, with part of speech, sense glosses, regional and register labels, inflected forms, and — the part that matters most here — explicit *countable* / *uncountable* / *plural-only* tagging. That single field resolves findings 3, 4 and 9 structurally instead of by hand-maintained exception lists, and its currency resolves finding 1.

USE FOR plural detection, spelling-variant merging, and modern vocabulary. Note the licence: Wiktionary is CC BY-SA, so anything you redistribute downstream inherits share-alike.

Kuperman age-of-acquisition norms 30,000 words · human-rated

Empirical mean age at which native speakers report learning each word, alongside Brysbaert's concreteness ratings for 40,000 lemmas. These are **the** answer to your 05Y / 10Y / 50Y tiers — measured from thousands of human raters rather than guessed by a language model, and stable, citable and reproducible in a way an LLM's judgement is not.

USE FOR the difficulty classification the project stalled on. Cite: Kuperman, Stadthagen-González & Brysbaert, *Behavior Research Methods* 44 (2012).

SUBTLEX-US frequency norms film and TV subtitles

Frequency counts drawn from spoken dialogue rather than written text. Word Chain is a spoken, fast, conversational game — subtitle frequency predicts what people actually reach for under a ten-second timer far better than book or web frequency does.

USE FOR ranking within tiers, and for the "is this a real word people say" filter that Wikipedia was doing badly.

wordfreq sunset September 2024 — and that is fine

The author stopped updating it, stating that generative AI had polluted the web text the counts were drawn from. For this project the frozen snapshot is arguably the *better* artefact: it captures human word usage from before the contamination. Pin the last release deliberately and say so in the dataset card.

USE FOR a second, pre-LLM frequency signal to cross-check SUBTLEX.

LisanBench public LLM word-chain benchmark

Directly relevant to your WCGT-AI proposal. It scores models on building long chains of English words where each step differs by one letter, measuring rule-following, vocabulary, planning, recall and persistence — the same five faculties you identified. It is *not* your game: last-letter chaining under a semantic category constraint is still unoccupied. But it means the framing "language games as an LLM benchmark" is now established rather than novel, and any WCGT-AI writeup has to position against it.

USE FOR prior art, scoring-metric design, and a reality check on the docx claim that "a kid can easily win" — that was written against GPT-4-1106 and needs re-running against 2026 reasoning models before it goes in a paper.

WHAT TO DO NEXT

A resumption plan

The strategic point: **the dataset's value was never its size**. Anyone can dump 41,466 nouns. What nobody has published is a noun list that knows how hard each word is, how often people say it, and what it does to a chain. Everything below builds toward that.

1 **Revoke the Wikimedia credentials and clean the docx**

Ten minutes, and it blocks nothing else. Move the server notes to a local file that never enters a shared project.

2 **Re-extract from Open English WordNet 2025 via wn**

Keep the old CSV as v0 for diffing — the delta between WordNet 3.0 and OEWN 2025 is itself a publishable observation about twenty years of English. Take the names split from Namenet instead of the capital-letter regex.

3 **Join against Wiktextextract on lemma**

Carry across countability, plural-only flags, regional spelling labels, and modern entries WordNet lacks. This is where

findings 1, 3, 4 and 9 get fixed at once. Keep every dropped entry as a labelled row, never a deletion.

4 **Attach WordNet supersenses, then AoA and frequency**

Supersense gives the BIO / MED / TEC / GEO axis for free. Kuperman AoA gives the 05Y / 10Y / 50Y axis empirically. SUBTLEX gives the ranking. Reserve the LLM classifier for the residue that none of the three cover — and grade it against the human norms first, so you can state its accuracy instead of trusting it.

5 **Compute the chain graph**

Twenty-six nodes, one directed edge per noun. Publish the 26×26 transition matrix, per-letter trap scores, and dead-end probabilities. Then the open question worth a paper: *what is the longest possible Word Chain?* It is a longest-trail problem on a directed multigraph — hard in general, tractable with this structure, and nobody has stated the answer for English nouns.

6 **Ship it as a versioned open dataset**

Hugging Face or Zenodo with a DOI, a dataset card documenting every filter as a reversible flag, and a licence that respects Wiktionary's share-alike. A citable artefact is what makes the two years of work count as research rather than a notebook.

7 **Then, and only then, WCGT-AI**

The benchmark needs the referee, and the referee needs the dataset — that dependency is why the benchmark stalled. Position it against LisanBench: your differentiator is the semantic category constraint and the difficulty tiers, which let you ask a question no one-letter-edit benchmark can, namely whether a model can stay inside a vocabulary band it was told to use.

Adjacent directions

- **The trap-score game mode.** Once every noun has a trap score, difficulty becomes a dial rather than a word list: an easy game bans -y endings, a hard one rewards them. That is a product feature no competing noun list can support.
- **Phonetic Word Chain.** The CMU Pronouncing Dictionary lets you chain on final *sound* rather than final letter — closer to how the game is played aloud, and it changes the whole letter economy (-y stops being a trap because it ends in a vowel sound). A second edge set over the same nouns, nearly free to compute.
- **The 20-year diff as a paper.** Extract from WordNet 3.0 and OEWN 2025 with an identical pipeline and publish what English gained and lost. You are one of very few people with the 2006 vintage